

PATRICK BROWN

Software Engineer

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EDUCATION

The University of Texas at Austin

Master of Science in Artificial Intelligence, GPA: 3.84

Fall 2025

Bachelor of Electrical and Computer Engineering, GPA: 3.87

May 2023

EXPERIENCE

H-E-B – Data Platform Engineer II ; Austin, TX

April 2025 – Present

- Architected a Kafka event bus unifying pipeline events from GCP Composer, Argo, and Databricks for cross-platform dependency tracking.
- Built a dependency service on the event bus for cross-platform pipeline orchestration and lineage.
- Owned end-to-end delivery from scoping through deployment; mentored junior engineers on cloud-native engineering.

H-E-B – Data Platform Engineer I ; Austin, TX

June 2023 – April 2025

- Designed, built, and deployed a full-stack internal product (React, FastAPI, PostgreSQL, Docker on AWS EKS) adopted by 10+ teams and 100+ users.
- Built a Confluence RAG assistant with autonomous re-embedding for internal search, before Atlassian shipped Roovo.

SELECTED PROJECTS

qwop-python – World-Record RL Environment

2026

- Beat the official HTML5 QWOP world record (45.167s vs. 45.530s) after transferring a Python-trained PPO policy to the real browser.
- Reverse-engineered minified QWOP JS into a headless Python Box2D gym; ~10,000 it/s per env (~100× vs. chromedriver), ~40,000 it/s in parallel.
- Ran a Grok Bot auto-research loop that killed plateaued policies, tuned hyperparameters, and shaped rewards.
- Turned the world-record run into a training objective: extracted keypresses from video, then a tunable gait metric after literal replay faceplanted.
- Closed the sim-to-browser gap with an opening fine-tune after Python policies faceplanted on official HTML5 physics.

Range Rat – Game

2026

- Shipped a golf game in the Godot engine using coding agents for gameplay logic; live at golf.bicrick.com.

MASTER'S IN AI

UT Austin MSAI Program

August 2024 – December 2025

- Built transformer language models from scratch in PyTorch (self-attention, causal masking) without pre-built attention layers.
- Implemented LoRA, 4-bit quantization, and QLoRA from scratch; built LLM generation and fine-tuning pipelines end-to-end.

ADDITIONAL INFORMATION

Languages: Python, TypeScript, JavaScript

Tools: PyTorch, Kafka, PostgreSQL, Terraform, Docker, Kubernetes, GCP, AWS, FastAPI, React

Skillsets: Pipeline orchestration, database systems, API layers, analytics dashboards

Honors: Eagle Scout (2018) · ACT 36 (2017)